

AGILE PROCESS MODEL DOCUMENT

Project Title

GramWatch – A Geo-Verified Community Grievance Reporting Platform for Rural India

1. Introduction

1.1 Purpose

This document describes the **Agile Process Model** adopted for the development of the GramWatch project. It explains the rationale for selecting Agile, the methodology followed, roles and responsibilities, artifacts, sprint planning, and execution strategy. This document is prepared as a formal process model submission for evaluation under the Software Engineering Lab.

1.2 Scope

The Agile Process Model covers the complete software development lifecycle of the GramWatch project—from requirement analysis to incremental development, testing, review, and refinement. It demonstrates how Agile practices ensure adaptability, continuous feedback, and structured progress tracking throughout the project duration.

2. Selection of Process Model

2.1 Chosen Model: Agile Scrum

The **Agile Scrum Model** is selected for the GramWatch project. Scrum is an iterative and incremental Agile framework that emphasizes flexibility, stakeholder feedback, and continuous improvement.

2.2 Justification for Choosing Agile

Agile Scrum is preferred over traditional models such as Waterfall because:

- Requirements may evolve based on user feedback from villagers and authorities
- The project involves UI-heavy and interaction-driven features
- Early delivery of working modules is beneficial
- Continuous testing and refinement improve reliability

- Agile supports academic projects with weekly deliverables

Thus, Agile Scrum aligns well with the technical and social nature of the GramWatch system.

3. Agile Principles Applied to GramWatch

The following Agile principles are applied:

- Incremental delivery of functional software
 - Continuous user and stakeholder feedback
 - Adaptability to requirement changes
 - Collaboration among team members
 - Emphasis on working software over documentation, while still maintaining necessary documents
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4. Scrum Roles and Responsibilities

4.1 Product Owner

- Represents system requirements and stakeholder interests
- Prioritizes features in the product backlog
- Ensures alignment with project objectives

4.2 Scrum Master

- Facilitates Agile practices
- Removes development obstacles
- Ensures adherence to Scrum framework

4.3 Development Team

- Designs, develops, and tests system features
 - Collaborates during sprint planning and reviews
 - Delivers incremental software builds
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5. Agile Artifacts

5.1 Product Backlog

The product backlog is a prioritized list of all desired system features.

Key Product Backlog Items:

- User registration and authentication
- Issue reporting with media upload
- GPS-based geo-tagging
- Community validation mechanism
- Validation threshold logic
- Escalation to Panchayat authorities
- Authority dashboard
- Issue status tracking
- Notification system

5.2 Sprint Backlog

The sprint backlog contains backlog items selected for a particular sprint along with implementation tasks.

5.3 Increment

Each sprint produces a functional increment of the GramWatch system that can be demonstrated and reviewed.

6. Sprint Planning and Execution

6.1 Sprint Duration

- Each sprint duration: **1 week**
- Total sprints planned: **5 sprints**

6.2 Sprint-wise Development Plan

Sprint 1: Requirement Analysis and Planning

- Study problem domain
- Requirement elicitation
- Preparation of SRS document
- Initial backlog creation

Sprint 2: System Design

- Architecture design
- UI wireframe preparation
- Database schema planning

Sprint 3: Core Feature Development

- User authentication module
- Issue reporting module
- GPS integration

Sprint 4: Validation and Escalation Logic

- Community validation feature
- Validation threshold controller
- Escalation workflow implementation

Sprint 5: Testing and Refinement

- Functional testing
 - Usability testing
 - Bug fixing and performance improvements
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7. Agile Ceremonies

7.1 Sprint Planning

- Selection of backlog items
- Task estimation and assignment

7.2 Daily Scrum

- Short daily meetings to discuss progress and obstacles

7.3 Sprint Review

- Demonstration of completed features
- Feedback collection

7.4 Sprint Retrospective

- Identification of process improvements
 - Discussion of challenges and solutions
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8. Advantages of Agile for GramWatch

- Faster delivery of usable features
 - Improved responsiveness to feedback
 - Better risk management through early testing
 - Enhanced transparency and progress tracking
 - Suitable for real-world, user-centric applications
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9. Comparison with Traditional Models

Aspect	Agile Model	Waterfall Model
Requirement Flexibility	High	Low
User Feedback	Continuous	Limited
Delivery	Incremental	Single final delivery
Risk Handling	Early	Late

Agile is therefore more suitable for the GramWatch project.

10. Conclusion

The Agile Scrum Process Model provides a structured yet flexible framework for developing the GramWatch system. By dividing development into manageable sprints and incorporating continuous feedback, the project ensures high quality, adaptability, and timely delivery. This process model supports both academic evaluation and real-world software engineering practices.